工程數學 Midterm #1 Oct 18, 2022

計算小錯誤無部份給分，請仔細作答

答案卷請務必在每一張寫上姓名、學號，沒寫不算分。Solve the ODEs:

Total 100 points; 10 points each.

1. $\frac{dy}{dx} = \frac{xy+2y-x-2}{xy-3y+x-3}$

2. $\frac{dy}{dx} + 2xy^2 = 0$

3. $\frac{dy}{dx} = y^2 - 4$

4. $y' + 2y = 0$ *has an integrating factor*(*積分因子*) e^{2x}, solve the ODE

5. $x\frac{dy}{dx} + 2y = 3$ *has an integrating factor*(*積分因子*) $I(x)$

   5-1. Find $I(x)$

   5-2. Solve the ODE:

6. $y'' - y' - 6y = 0$

7. $y^{<4>} - 2y'' + y = 0$

8. $y'' + 16y = 0, \quad y(0) = 2, \quad y'(0) = -2$

9. $y'' - y' = -3$

10. $x^2y'' - 2y = 0$

公式表

求 y_h:

Auxiliary equation: $m^2 + am + b = 0$　　根為: $\frac{-a \pm \sqrt{a^2-4b}}{2}$

$\sqrt{a^2-4b}$*解有三種可能*:

<1>相異實根: m_1, m_2

ODE 通解為: $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

<2>實數重根: m

ODE 通解為: $y = (c_1 + c_2 x) e^{mx}$

<3>複數根 (共軛虛根): $\alpha \pm \omega i$

ODE 通解為: $y = e^{\alpha x}(A \cos \omega x + B \sin \omega x)$

求 y_p:

Undetermined coefficients method (未定係數法):

R(x)	y_p 假設型
k	A
e^{ax}	Ae^{ax}
cos bx 或 sin bx	A cos bx + B sin bx
x^n	$A_n x^n + A_{n-1} x^{n-1} + \ldots + A_1 x^1 + A_0$
cx^n	$A_n x^n + A_{n-1} x^{n-1} + \ldots + A_1 x^1 + A_0$
$x^n e^{nx}$	$e^{nx}(A_n x^n + A_{n-1} x^{n-1} + \ldots + A_1 x^1 + A_0)$
$cx^n e^{nx}$	$e^{nx}(A_n x^n + A_{n-1} x^{n-1} + \ldots + A_1 x^1 + A_0)$
$x^n \cos bx$ 或 $x^n \sin bx$	$(A_n x^n + A_{n-1} x^{n-1} + \ldots + A_1 x^1 + A_0) \cos bx$ + $(B_n x^n + B_{n-1} x^{n-1} + \ldots + B_1 x^1 + B_0) \sin bx$

註: a, b, c, k, n, A, B 為常數

- Cauchy-Euler equation:

  令 t =ln x (以 t 取代 x)

  $xy' = D_t y$

  $x^2 y'' = D_t(D_t - 1)y$

  $x^3 y''' = D_t(D_t - 1)(D_t - 2)y$